

Worksheet

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1. (The idea for this exercise is from Liza Bolton.) The following code makes a nice graph. Modify it to create the ugliest graph that you can.

```
library(ggplot2)

ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width, colour = Species)) +
  geom_point(size = 2.5, alpha = 0.8) +
  labs(
    title = "Iris sepal dimensions",
    x = "Sepal length (cm)",
    y = "Sepal width (cm)",
    colour = "Species"
  ) +
  theme_minimal()
```

2. In groups of two, pick one of this week's readings and make a two-to-five slide summary of its most important points using Quarto. Make a PR that adds your slides to the class GitHub repo.
 - Timbers et al., *Reproducible and Trustworthy Workflows for Data Science*, the version control chapters.
 - Polars User Guide, "Getting started" and "Concepts", together with the Parquet page of the Apache Arrow Python docs.
 - Healy, *Data Visualization*, Chapter 1.
 - Alexander, *Telling Stories with Data*, Chapter 5.
3. Make a Quarto website and put it online. Create the project (`quarto create project website`), add a page with one graph of the shelter data from Week 1, set `output-dir: docs` in `_quarto.yml`, render, and push to a new GitHub repository. Under Settings, then Pages, deploy from `main` and the `/docs` folder. Write down the URL and open it on someone else's laptop.
4. Make three graphs from three different Open Data Toronto datasets using `ggplot2`, and one table using `tinytable`, and add them as a page on your website.